

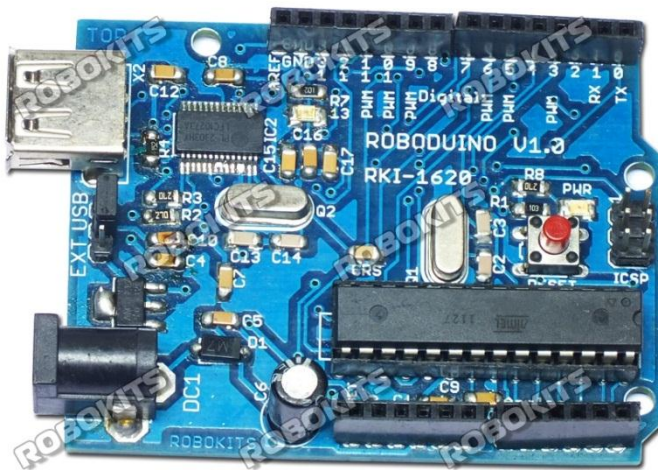
ELP305: Expt No. 0 - Introduction to Arduino

Introduction To Arduino:

Arduino is an open source electronics prototyping platform based on flexible, easy to use hardware and software. Arduino can sense the environment by receiving input from a variety of sensors and can act its surroundings by controlling lights, motors and other actuators. The microcontroller on the board is programmed using the Arduino programming language (based on Wiring) and the Arduino development environment (based on Processing). Arduino is composed of two major parts: the Arduino board, which is the piece of hardware we work on when you build your objects, and the Arduino IDE, the piece of software we run on our computer. We use the IDE to create a sketch (a little computer program) that you upload to the Arduino board. The sketch tells the board what to do.

Hardware

Arduino is a single-board microcontroller. The hardware consists of an open-source hardware board designed around an 8-bit Atmel AVR microcontroller, or a 32-bit Atmel ARM with complementary components to facilitate programming and incorporation into other circuits. Arduino has total 20 I/O pins.



Objective - LED Flasher

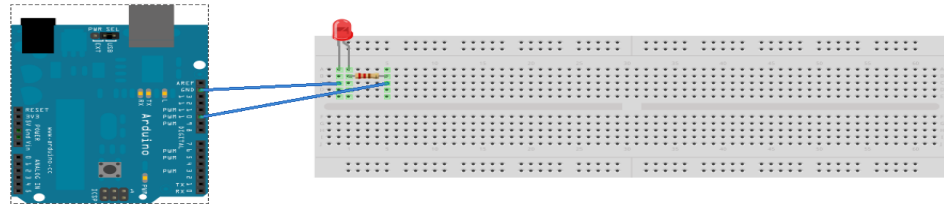
Stuff you will need to have:

1. Arduino Board - www.arduino.cc
2. Bread Board
3. LED
4. 100 ohms Resistor
5. Jumper Wire

Experimental setup

For the first project, we are going to connect an LED to one of the digital I/O pins of Arduino. We control the blinking of LED with Arduino. First, make sure your Arduino is powered off by unplugging it from the USB cable. Now, take your breadboard, LED, resistor, and wires and connect everything.

Schematic



LED Flasher

Made with  Fritzing.org

Going Further:

1. Interfacing with seven segment LED to make a counter using a push button as a input.
2. Displaying different patterns on LED matrix.

References:

1. <http://www.arduino.cc/>
2. All about Arduino hardware - <http://www.allaboutcircuits.com/technical-articles/understanding-arduino-uno-hardware-design/>
3. Board specifications: <https://www.arduino.cc/en/Guide/Board?from=Tutorial.ArduinoBoard>
4. Getting Started with Arduino: <http://www.makeuseof.com/tag/started-arduino-starter-kit-installing-drivers-setting-board-port/>
5. Schematic and I/O pins for Arduino :<https://eelabs.wordpress.com/2014/04/14/schematic-and-inputoutput-pins-explain-for-arduino-uno-r3/>
6. <http://arduino.cc/en/Tutorial/Sketch>
7. Basics for Fritzing : <http://www.instructables.com/id/Fritzing-A-Tutorial/?ALLSTEPS>
8. <http://fritzing.org/learning/tutorials>